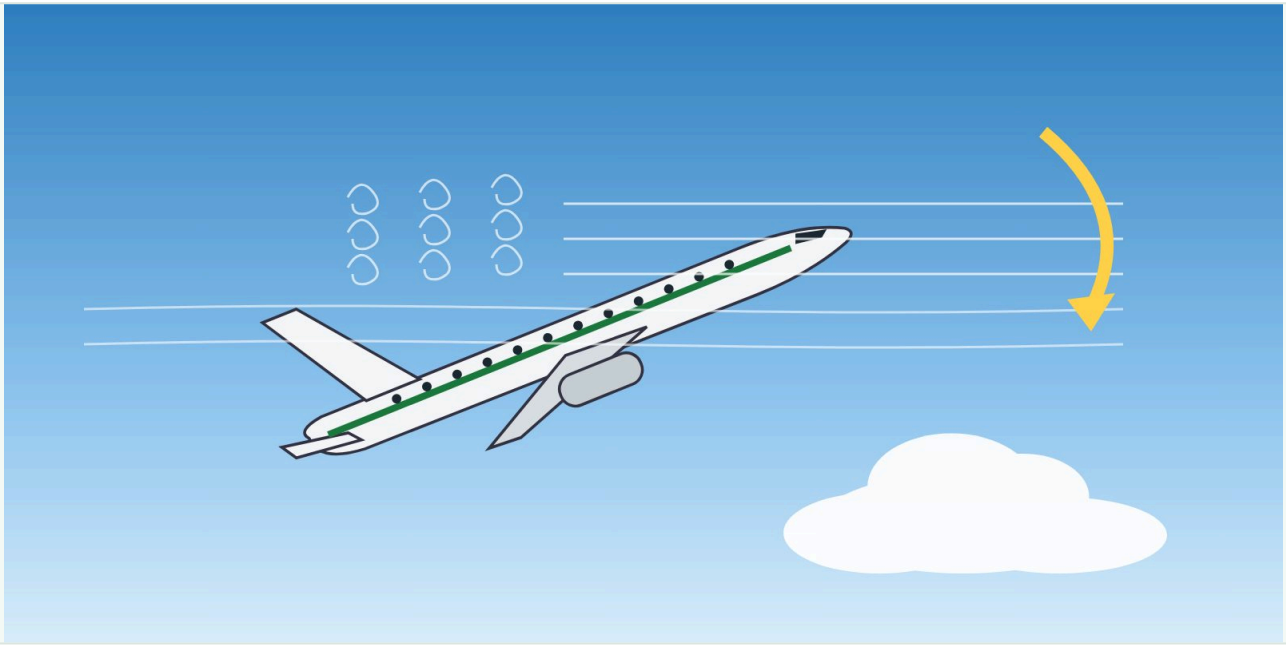
MODULE 6 · RECOVERY

Lesson 20: Stall Recovery

Lesson 20 of 24 · about 8 min

LEARNING OBJECTIVES

- Apply the industry stall recovery template.
- Explain each step and the reasons behind it.
- Recognise the conditions that make stall recovery difficult.



Stall recovery: reduce angle of attack first.

The stall recovery template

Following accidents in which crews tried to "power out" of stalls while holding altitude, manufacturers and authorities adopted a common stall recovery template. A typical version is shown below. **Use your manufacturer's procedure.**

Step	Action	Why
1	**Autopilot and autothrust — disconnect**	The automation may be holding the nose up or reducing thrust; it will not recover a stall.
2a	**Nose-down pitch control — apply until the stall warning stops** and the aircraft is no longer stalled	Reducing the angle of attack is the only way to unstall the wing.
2b	**Nose-down pitch trim — as needed**	If the aircraft was trimmed nose-up, the elevator alone may not be enough.
3	**Bank — wings level**	Levelling the wings directs the lift vector upwards and reduces the load factor.

4	**Thrust — as needed**	Add thrust once the angle of attack is reduced; be aware of the nose-up pitching moment of underwing engines. At low speed with high thrust, reducing thrust may be needed to lower the nose.
5	**Speedbrakes / spoilers — retract**	They increase drag and the stall speed.
6	**Return to the desired flight path**	Recover smoothly, avoiding a secondary stall warning or excessive g.

Important points

- **Reducing angle of attack comes before everything else — including altitude.** Accept the altitude loss.
- **The stall warning is a stall.** Treat any stall warning (except a confirmed false warning on take-off, if your procedure allows) as requiring the recovery template.
- A nose-down input may need to be **large and sustained**. A pitch attitude below the horizon may be required.
- Avoid a **secondary stall**: do not pull up too quickly after the warning stops.
- If the stick pusher activates, **do not oppose it**.
- Near the ground, the same template applies; only the urgency changes.

When stall recovery becomes difficult

- High altitude — little excess thrust, large altitude loss needed (Lesson 7).
- Nose-up trim — the stabiliser may need to be trimmed nose-down.
- High thrust with underwing engines — the nose-up moment opposes the recovery.
- Startle and unreliable instruments — the crew may not recognise that the aircraft is stalled.

KEY TAKEAWAYS

- Template: automation off → nose down (and trim) until unstalled → wings level → thrust as needed → speedbrakes retracted → return to flight path.
- Angle of attack first — altitude is secondary.
- Avoid a secondary stall; never oppose the stick pusher.

KNOWLEDGE CHECK

During stall recovery, the nose-down elevator input does not reduce the angle of attack enough. What else may be needed?

A. Nose-up trim

B. Nose-down pitch trim, and reducing thrust if high thrust is causing a nose-up moment

C. Full rudder

D. Extending the speedbrakes

Answer: B. If the aircraft is trimmed nose-up or high underwing thrust is pitching it up, nose-down trim and a thrust reduction may be needed to reduce the angle of attack.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.